

Personalized treatment strategies for breast adenoid cystic carcinoma: A machine learning approach

Sakhr Alshwayyat^{a,b,c}, Mahmoud Bashar Abu Al Hawa^d, Mustafa Alshwayyat^d,
Tala Abdulsalam Alshwayyat^b, Siya sawan^e, Ghaith Heilat^f, Hanan M. Hammouri^g,
Sara Mheid^h, Batool Al Shweiatⁱ, Hamdah Hanifa^{j,*}

^a King Hussein Cancer Center, Amman, Jordan

^b Princess Basma Teaching Hospital, Irbid, Jordan

^c Research Fellow, Applied Science Research Center, Applied Science Private University, Amman, Jordan

^d Faculty of Medicine, Jordan University of Science & Technology, Irbid, Jordan

^e Faculty of Medicine, University of Jordan, Amman, Jordan

^f Breast Oncoplastic and General Surgery, Department of General Surgery and Urology, Jordan University of Science & Technology, King Abdullah University Hospital, Irbid, Jordan

^g Department of Mathematics and Statistics, Faculty of Arts and Science, Jordan University of Science and Technology, Irbid, Jordan

^h Radiation Oncology Department, King Hussein Cancer Center, Amman, Jordan

ⁱ Breast Imaging Fellow, Department of Radiology, King Hussein Cancer Center, Amman, Jordan

^j Faculty of Medicine, University of Kalamoon, Al Nabk, Syria

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ABSTRACT

Background: Breast adenoid cystic carcinoma (BACC) is a rare subtype of breast cancer that accounts for less than 0.1 % of all cases. This study was designed to assess the efficacy of various treatment approaches for BACC and to create the first web-based tool to facilitate personalized treatment decisions.

Methods: The Surveillance, Epidemiology, and End Results (SEER) database was used for this study's analysis. To identify the prognostic variables, we conducted Cox regression analysis and constructed prognostic models using five Machine Learning (ML) algorithms to predict the 5-year survival. A validation method incorporating the area under the curve (AUC) of the receiver operating characteristic (ROC) curve was used to validate the accuracy and reliability of ML models. We also performed a Kaplan-Meier (K-M) survival analysis.

Results: This study included 1212 patients. The median age was 60 years, with most tumors being localized and less than 2 cm in size. The 5-year overall survival (OS) rates were highest for surgery + radiotherapy (RT) (94.9 %) and lowest for surgery + chemotherapy (CTX) + RT (80.1 %). Positive estrogen receptor (ER) status and younger age were associated with better survival outcomes. ML models identified key predictive features for survival, including age, nodal status, and ER status.

Conclusion: Age, lymph node metastasis, and ER status are crucial prognostic indicators for BACC. Although postoperative RT enhances survival, the advantages of adjuvant CTX are uncertain, implying that it may be eschewed to avert adverse effects. Our online tool offers essential resources for prognostication and treatment optimization.

1. Introduction

Breast cancer is the most prevalent type of malignancy and a leading cause of cancer-related death in women worldwide [1]. Among the

various types of breast cancer, adenoid cystic carcinoma (ACC) is a rare subtype, accounting for less than 0.1 % of reported cases [2].

BACC is associated with better outcomes than extramammary ACC, with 10-year survival rates ranging from 85 % to 100 %. In comparison,

* Corresponding author.

E-mail addresses: sakhrabdusalam@gmail.com (S. Alshwayyat), mahmoud2000hawa@gmail.com (M.B. Abu Al Hawa), maalshwayyat22@med.just.edu.jo (M. Alshwayyat), tala.abdelsalam@yahoo.com (T.A. Alshwayyat), Siya.sawan.7@gmail.com (S. sawan), gbheilat@just.edu.jo (G. Heilat), hmhammouri@just.edu.jo (H.M. Hammouri), SM.11445@khcc.jo (S. Mheid), Ba.17390@khcc.jo (B. Al Shweiat), hamdahhanifa@gmail.com (H. Hanifa).

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patients with ACC in the head and neck region had a survival rate of 30–54 % [3]. Although local recurrence and distant metastasis are common within ten years, the lung is the most common site of metastasis [4]. Notably, metastases may emerge many years after the initial diagnosis, and several studies have reported prolonged survival rates [5].

Late local recurrences, which occur years after the initial diagnosis, are relatively frequent, with a 5-year local recurrence rate of 6 % and a 10-year rate of 31 % [6]. Fortunately, most recurrences can be treated effectively, and the long-term prognosis remains favorable [6,7]. Given the delayed onset of local and distant recurrence, long-term follow-up is essential for such patients [4]. The scarcity of data owing to the rarity of BACC results in a significant knowledge gap, making it challenging to establish standardized treatment protocols [2].

Surgery is the primary treatment for BACC, but the rarity of this cancer type leads to variations in treatment strategies due to the lack of clear guidance on the appropriate surgical approach [8]. For grade I tumors, tumor lumpectomy is recommended, whereas mastectomy is recommended for grade II tumors. Mastectomy combined with lymph node dissection has been suggested for grade III tumors [9]. It is suggested that breast-conserving treatments, including postoperative RT, are as effective as mastectomy alone in terms of survival; however, mastectomy may be considered for large tumors given the successful salvage of all local failures [10].

Studies have shown that postoperative adjuvant RT can improve overall and disease-specific survival in patients who have undergone local surgery [11]. Therefore, postoperative RT should be offered to all patients following lumpectomy, regardless of the margin status [10].

Despite this, there is still considerable debate regarding the use of adjuvant CTX after surgery, and no definitive consensus has been reached. Some researchers argue that adjuvant CTX does not improve disease-free or OS in patients with breast cancer [12]. Unadjusted local and distant recurrence rates were notable, indicating that adjuvant CTX and radiotherapy may be necessary in specific cases [13]. Late recurrences can occur; therefore, long-term follow-up is required [2].

The limited understanding of BACC owing to its rarity highlights a significant knowledge gap, particularly in identifying effective treatment strategies and prognostic factors. Conventional methods for studying this type of cancer are constrained by their small sample sizes and limited clinical data. In recent years, the integration of artificial intelligence (AI) and ML has shown promise in oncology, offering the ability to analyze complex datasets and uncover patterns that might otherwise remain undetected [14].

In breast cancer, ML has become the cornerstone of personalized medicine through molecular subtype characterization, diagnostic accuracy, and treatment response prediction. Techniques such as deep neural networks and neural networks have been used for tumor detection, lymph node identification, and molecular profiling, significantly enhancing precision and clinical decision-making [15].

Moreover, high-throughput omics technologies analyzed using ML algorithms have identified intrinsic molecular subtypes, aiding in prognosis and treatment planning [15]. Beyond molecular insights, frameworks such as MOSAIC integrate clinical and genomic features to improve survival predictions and patient stratification, particularly in rare cancers, emphasizing the adaptability of ML to the unique challenges posed by diseases such as BACC [16].

Some of them, such as OncoNPC, use next-generation sequencing and electronic health records to predict tumor origins, with some cases reaching complete accuracy for cancers of unknown primary origin [17]. Previous applications of ML in rare tumors have shown its potential for uncovering new prognostic factors, optimizing therapeutic strategies, and improving survival rates [14,18–22]. These advances underline the role of ML in solving some of the challenges presented by infrequent subtypes, such as BACC, for which classical approaches mostly fail.

In this context, our study aimed to address these gaps by employing ML-based techniques to evaluate the prognostic value of various

treatment modalities and the impact of clinicopathological factors on survival. By leveraging interactive AI models, we seek to provide data-driven insights that will guide clinicians in making personalized and evidence-based treatment decisions for BACC while laying the groundwork for future research in this underexplored domain.

2. Methodology

2.1. Data extraction and variables

We extracted data from the Surveillance, Epidemiology, and End Results (SEER) database supported by the National Cancer Institute, which comprises approximately 27.8% of the US population. Patients diagnosed with ACC based on the third edition of the International Classification of Diseases for Oncology (ICD-O-3) were included in this study [23]. The histological code for the disorder was 8200/3 (adenoid cystic carcinoma). Patients with the following criteria were excluded: a diagnosis not confirmed by histology, not the first tumor or presence of other malignancies in the body, and unknown data. Finally, 1212 eligible patients were included. The detailed screening process is shown in the flow diagram in Fig. 1.

The following SEER variables were selected for our study: patient characteristics (age, race, marital status, and year of diagnosis), tumor characteristics (laterality, primary site, tumor size, SEER stage, metastasis, hormonal status, AJCC stage, T and N of AJCC component), therapeutic methods (surgery, RT, and CTX), and survival information obtained from the database. Age was divided into <60 years and >60 years. Tumor size was categorized into two groups: <2 and >2 cm; and SEER stages were classified into three groups: localized (within organ), regional (extension to adjacent organs or regional lymph nodes), and distant (involvement of distant organs or distant metastasis). The primary survival outcomes were OS, calculated from diagnosis to any cause of death, and cancer-specific survival (CSS), calculated from diagnosis to death due to BACC.

2.2. Data processing and statistical analysis

All the data were analyzed using R software (v4.2.2) and the following packages: "readxl," "tidyverse," "Hmisc," "data.table," "Table 1," "MatchIt," "survminer," "survival," and "broom." Chi-square analysis was used to assess categorical variables. The K-M method was used to determine survival rates, and the log-rank test analyzed differences in survival curves. Cox regression models were used to identify independent predictors of OS and CSS. Statistical significance was set at a two-tailed P-value of less than 0.05.

Five different machine learning models (MLM) were applied in this study to predict 5-year survival rates [24]. To increase accuracy, Random Forest Classifier (RFC) was used to create several decision trees and combine their results. Gradient Boosting Classifier (GBC) to build trees sequentially, with each tree's estimation error correcting the previous one. Based on patient characteristics, Logistic Regression (LR) was used to estimate the probability of survival by using statistical approaches. K-Nearest Neighbors (KNN) to predict survival by comparing patients with the most similar patients in the dataset. Multilayer Perceptron (MLP), a type of neural network, to learn from data by adjusting the connections between layers of "neurons." These models helped us understand the complex relationships between patient characteristics and survival outcomes, providing valuable insights into prognosis.

2.3. Model training and evaluation

Standard scaling was applied uniformly across all features to ensure model robustness using the StandardScaler module of Scikit-learn. This normalizes the data, alleviating the potential impacts of different scales among the features. Subsequently, ML models were trained on a dataset with a binary classification output predicting the target "5-year

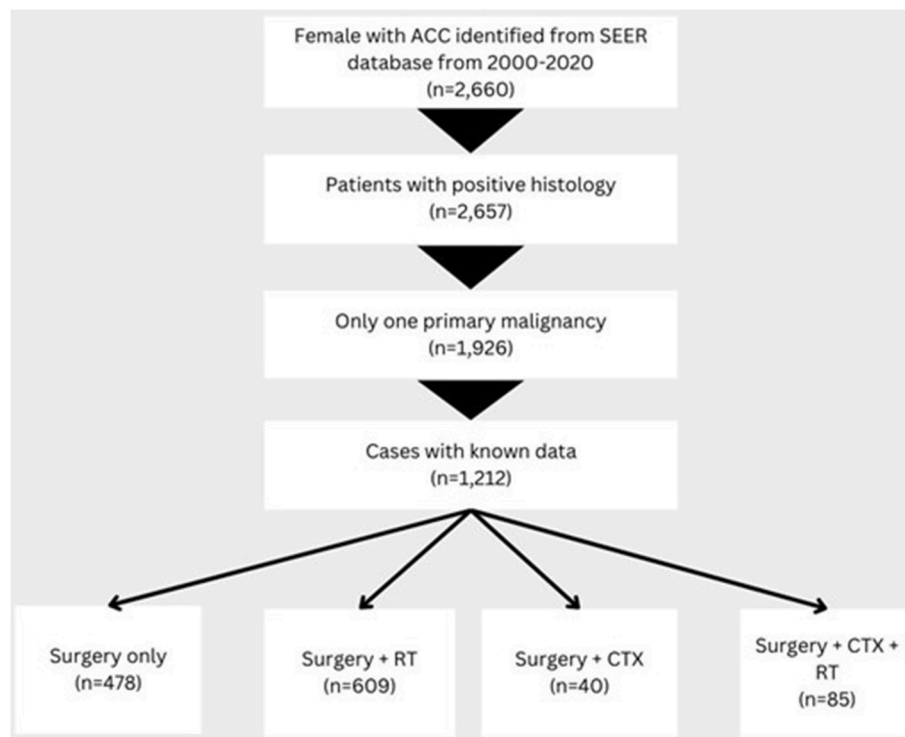


Fig. 1. Flow diagram for patients' selection.

survival." The features included significant variables in the multivariate regression and therapeutic options. The dataset was randomly divided into 8:2 training ($n = 970$) and testing ($n = 242$) sets.

The feature contribution to predicting "5-year survival" was calculated using the permutation importance method. ROC curves and AUC scores were used to evaluate the model's discriminatory power. The `roc_curve` function from the `sklearn.metrics` module computes False Positive Rate (FPR) and True Positive Rate (TPR). The predicted probabilities of the positive class were obtained using the `predict_proba` method for each model. AUC scores were calculated using the `roc_auc_score` function.

A custom plotting function, `plot_roc_curve`, was used to visualize the ROC curves of multiple models. Additional model evaluations included a Mean Bootstrap Estimate with a 95 % Confidence Interval, 10-fold cross-validation, and a classification report for precision, recall, and F1-score. All ML implementations were processed using the `scikit-learn` 0.18 package in Python.

3. Results

3.1. Clinicopathological characteristics

The study population comprised 1212 patients. Among them, 47.9 % were <60 years of age, and the median age was 60 years. A total of 52.2 % of the patients had a tumor size <2 cm, with a median tumor size of 1.9 cm. The largest racial group was white, comprising 79.5 % of the cases, and 12.1 % of the cases were black. Furthermore, most cases were localized (95.1 %), T1 (56.2 %), and N0 (96.4 %). All cases without distant metastasis (M0). The most common type of surgery was partial mastectomy (74.8 %), followed by total mastectomy without removal of uninvolved contralateral breast (14.6 %). The demographic and clinical characteristics of patients with BACC are shown in Table 1.

3.2. Survival analysis

As shown in Fig. 2, the therapeutic groups showed significant

differences in OS and CSS ($P < 0.001$). The survival rates for different therapeutic options were analyzed: surgery + RT showed a 94.9 % 5-year OS rate, surgery only exhibited a survival rate of 91 %, surgery + CTX showed an 83.5 % rate, and surgery + CTX + RT was 80.1 %. In terms of CSS, the surgery + RT, surgery only, surgery + CTX, and surgery + CTX + RT groups had survival rates of 98.2 %, 97.1 %, 83.5 %, and 83 %, respectively.

3.3. Prognostic factors

To determine the possible independent prognostic factors for OS and CSS in BACC, we adopted a Univariate Cox regression model and selected only the significant variables for multivariate Cox regression. Older age and N1 were a poor prognostic factor for OS, whereas positive ER status was a good prognostic factor for OS Fig. 3. In terms of CSS, lower inner and upper outer quadrants were a good prognostic factor, whereas N1 was associated with a worse prognosis.

3.4. Model performances and interpretability

The detailed performance metrics for all the ML algorithms are summarized in Table 2. ROC curves of all ML models are displayed in Fig. 4A. The features that contributed the most to the survival prediction are displayed in Fig. 4B. The highest contributing factors were age, followed by N component of AJCC stage, and ER status. A web-based tool was developed using a random forest model that incorporated various clinical variables to predict the survival of patients diagnosed with BACC. Clinicians and researchers can input patient-specific data to estimate survival duration.

The interactive tool, designed for ease of use and rapid calculation, is accessible through the following link: <https://sakhralshwayyat.shinyapps.io/AdenoidCysticCarcinoma/>. For instance, the tool predicts the expected survival months upon entering patient data.

Table 1
Clinicopathological characteristics.

Category Characteristics	Surgery (N = 478)	Surgery + CTX (N = 40)	Surgery + CTX + RT (N = 85)	Surgery + RT (N = 609)	Overall (N = 1212)	P-value
Age						
<60 years	204 (42.7 %)	34 (85.0 %)	55 (64.7 %)	287 (47.1 %)	580 (47.9 %)	<0.001
>60 years	274 (57.3 %)	6 (15.0 %)	30 (35.3 %)	322 (52.9 %)	632 (52.1 %)	
Race						
Asian or Pacific Islander	61 (12.8 %)	4 (10.0 %)	4 (4.7 %)	33 (5.4 %)	102 (8.4 %)	0.002
Black	63 (13.2 %)	3 (7.5 %)	13 (15.3 %)	68 (11.2 %)	147 (12.1 %)	
White	354 (74.1 %)	33 (82.5 %)	68 (80.0 %)	508 (83.4 %)	963 (79.5 %)	
Primary Site						
The central portion of the breast	45 (9.4 %)	3 (7.5 %)	16 (18.8 %)	33 (5.4 %)	97 (8.0 %)	0.002
Upper-inner quadrant of breast	49 (10.3 %)	0 (0 %)	0 (0 %)	54 (8.9 %)	103 (8.5 %)	
Lower-inner quadrant of breast	69 (14.4 %)	9 (22.5 %)	13 (15.3 %)	75 (12.3 %)	166 (13.7 %)	
Upper-outer quadrant of the breast	160 (33.5 %)	12 (30.0 %)	26 (30.6 %)	235 (38.6 %)	433 (35.7 %)	
Lower-outer quadrant of the breast	51 (10.7 %)	6 (15.0 %)	9 (10.6 %)	52 (8.5 %)	118 (9.7 %)	
Overlapping lesion of the breast	104 (21.8 %)	10 (25.0 %)	21 (24.7 %)	160 (26.3 %)	295 (24.3 %)	
Laterality						
Left - the origin of the primary	233 (48.7 %)	28 (70.0 %)	42 (49.4 %)	297 (48.8 %)	600 (49.5 %)	0.138
Right - the origin of the primary	245 (51.3 %)	12 (30.0 %)	43 (50.6 %)	312 (51.2 %)	612 (50.5 %)	
Marital Status						
Married	266 (55.6 %)	28 (70.0 %)	56 (65.9 %)	362 (59.4 %)	712 (58.7 %)	0.207
Other	212 (44.4 %)	12 (30.0 %)	29 (34.1 %)	247 (40.6 %)	500 (41.3 %)	
PR Status						
Negative	427 (89.3 %)	31 (77.5 %)	80 (94.1 %)	552 (90.6 %)	1090 (89.9 %)	0.061
Positive	51 (10.7 %)	9 (22.5 %)	5 (5.9 %)	57 (9.4 %)	122 (10.1 %)	
ER Status						
Negative	356 (74.5 %)	31 (77.5 %)	74 (87.1 %)	485 (79.6 %)	946 (78.1 %)	0.075
Positive	122 (25.5 %)	9 (22.5 %)	11 (12.9 %)	124 (20.4 %)	266 (21.9 %)	
Tumor size						
<2 cm	257 (53.8 %)	9 (22.5 %)	27 (31.8 %)	340 (55.8 %)	633 (52.2 %)	<0.001
>2 cm	221 (46.2 %)	31 (77.5 %)	58 (68.2 %)	269 (44.2 %)	579 (47.8 %)	
Stage						
Localized	463 (96.9 %)	36 (90.0 %)	54 (63.5 %)	600 (98.5 %)	1153 (95.1 %)	<0.001
Regional	15 (3.1 %)	4 (10.0 %)	31 (36.5 %)	9 (1.5 %)	59 (4.9 %)	
T						
T1	278 (58.2 %)	15 (37.5 %)	33 (38.8 %)	355 (58.3 %)	681 (56.2 %)	<0.001
T2	188 (39.3 %)	21 (52.5 %)	52 (61.2 %)	223 (36.6 %)	484 (39.9 %)	
T3	12 (2.5 %)	4 (10.0 %)	0 (0 %)	31 (5.1 %)	47 (3.9 %)	
N						
N0	471 (98.5 %)	37 (92.5 %)	54 (63.5 %)	606 (99.5 %)	1168 (96.4 %)	<0.001
N1	7 (1.5 %)	3 (7.5 %)	31 (36.5 %)	3 (0.5 %)	44 (3.6 %)	
AJCC Stage						
I	278 (58.2 %)	12 (30.0 %)	30 (35.3 %)	355 (58.3 %)	675 (55.7 %)	<0.001
II	200 (41.8 %)	28 (70.0 %)	55 (64.7 %)	254 (41.7 %)	537 (44.3 %)	
Metastasis						
M0	478 (100 %)	40 (100 %)	85 (100 %)	609 (100 %)	1212 (100 %)	<0.001
Surgery Type						
Modified radical mastectomy without removal of uninvolved contralateral breast	40 (8.4 %)	6 (15.0 %)	9 (10.6 %)	9 (1.5 %)	64 (5.3 %)	<0.001
Partial mastectomy	244 (51.0 %)	12 (30.0 %)	76 (89.4 %)	574 (94.3 %)	906 (74.8 %)	
Total mastectomy with removal of uninvolved contralateral breast	53 (11.1 %)	12 (30.0 %)	0 (0 %)	0 (0 %)	65 (5.4 %)	
Total mastectomy without removal of uninvolved contralateral breast	141 (29.5 %)	10 (25.0 %)	0 (0 %)	26 (4.3 %)	177 (14.6 %)	

4. Discussion

BACC resembles a salivary gland characterized by minimal nodal involvement, late distant metastasis, single lesions, and excellent prognosis. Although predominantly triple-negative, it is generally considered indolent [25,26]. A population-based study in the United States revealed a higher incidence of BACC among white individuals than among other races [26], a trend that was also observed in our study, where 79.5 % of the patients were white. The rarity of this condition leads to variability in treatment approaches. Using ML, we investigated the prognostic impact of multiple variables and the survival value of various treatment methods to guide future research and contribute to the development of personalized medicine.

ML models, including Gradient Boosting Survival and frameworks such as MOSAIC, show outstanding predictive performance in the prognosis of survival and recurrence for cancers using clinical and genomic data for fine stratification of patients [14,16]. In rare pediatric

cancers, AI-enabled predictive modeling has advanced personalized treatment approaches by incorporating multimodal datasets and overcoming the challenges of small sample size and heterogeneity [16]. According to Kaliappan et al. (2023) [27], this is the case here. However, according to Sun (2024) [28], the application of ML in rare cancers is limited by small datasets that restrict model training and result in overfitting. Large interpretability and bias in deep neural networks due to imbalance in data-a basic hurdle for clinical acceptance are the need of the hour innovations such as explainable AI and federated learning [29,30]. Despite this, this study shows how ML optimizes treatment planning and helps attain better survival in BACC through robust modeling techniques, thereby addressing these shortcomings.

Our study found that older age had a detrimental impact on survival, especially in patients over 60 years of age. Another study corroborated our findings, reporting that the age group 45–59 showed the best OS and CSS rates among four age subgroups (>44, 60–75, 75+) [31]. This may be due to the poorer health of older patients and their reduced tolerance

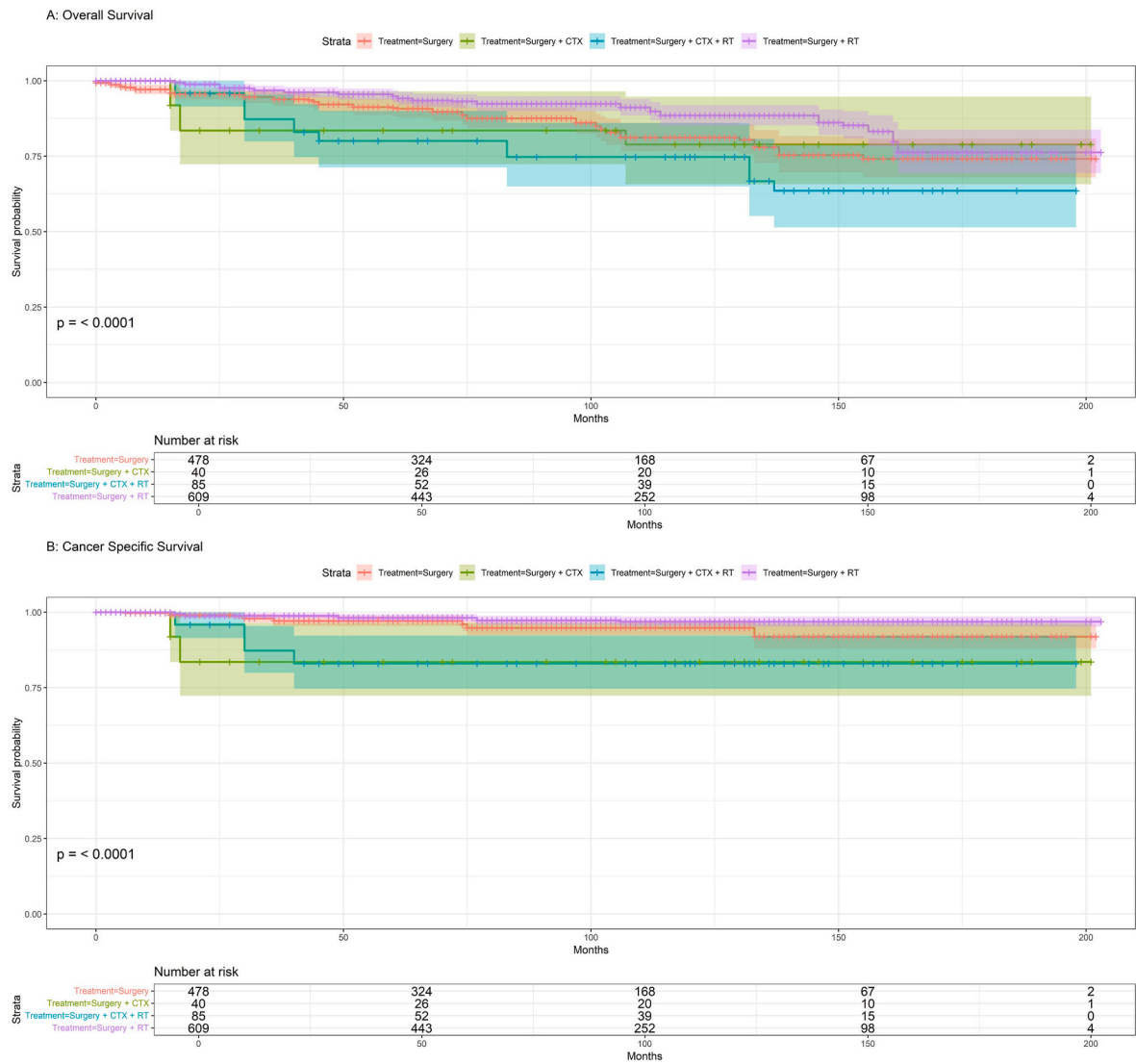


Fig. 2. a) Kaplan-Meier overall survival (OS) for treatment; b) Kaplan-Meier cancer-specific survival (CSS) for treatment.

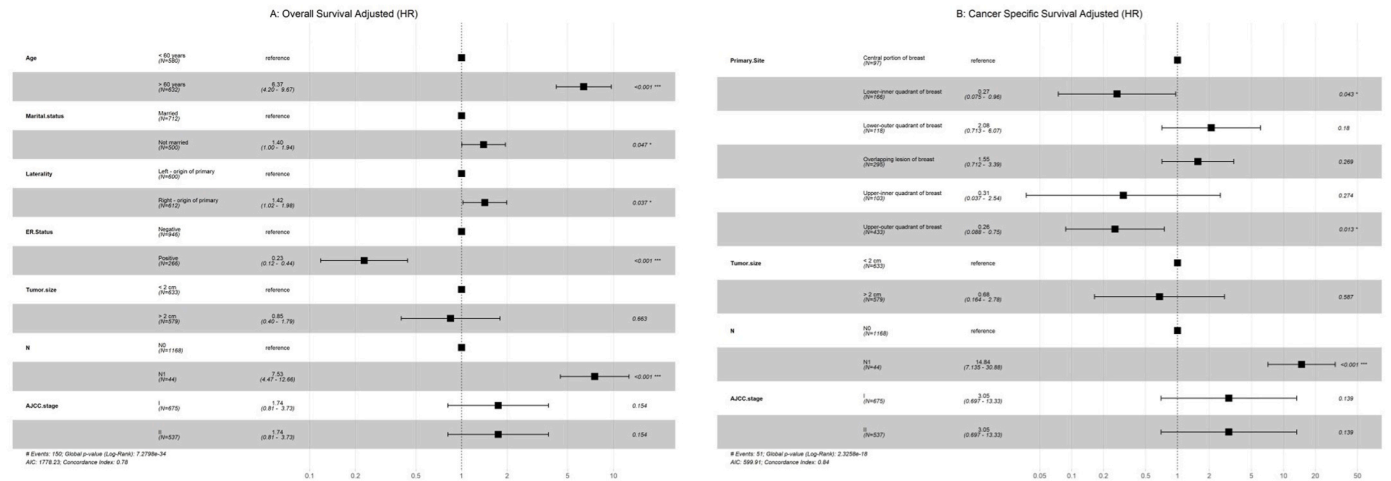


Table 2
Machine learning algorithms performance.

Model	Accuracy	Precision	Recall	F1 score	AUC
Logistic Regression	68.95 %	69.50 %	95.21 %	80.35 %	0.6820
K-Nearest Neighbors	64.84 %	77.17 %	67.12 %	71.79 %	0.6970
Random Forest Classifier	70.78 %	71.13 %	94.52 %	81.18 %	0.7430
Gradient Boosting Classifier	70.78 %	71.13 %	94.52 %	81.18 %	0.7340
MLP Classifier	70.32 %	70.98 %	93.84 %	80.83 %	0.7430

to aggressive treatment. Therefore, age should be considered a significant factor in predicting prognosis and selecting treatments.

The ER status of the disease is almost always negative [26]; however, some studies have reported positive ER status [32–34]. Our study found that 78.1 % of all patients had a negative ER status, and our analysis indicated that a positive ER status was a favorable prognostic factor. This contrasts with multiple studies that reported no significant effect of a positive ER status [35,36]. The potential role of hormonal therapy in the treatment of these patients remains unclear and warrants further investigation.

Axillary nodal metastasis is rare in BACC, although higher rates have been reported in the solid basaloid type [37]. Multiple studies have reported very low rates of axillary lymph node involvement [38,39]. A 20-year retrospective study showed that none of the six identified

patients had nodal involvement, suggesting that node excision may be unnecessary without debilitating factors, such as concurrent cancer or high-grade lesions [40]. In our study, 5.3 % of the patients underwent radical mastectomy, which may be indicated regardless of nodal involvement due to evidence of recurrence and deep invasion without axillary metastases [32]. The low rate of such surgery likely reflects the rarity and indolent nature of BACC. Additionally, N1 nodal status was a poor prognostic factor in our Cox regression model, indicating that further testing to identify axillary involvement is warranted even with low incidence rates.

The general treatment for BACC is surgical excision, although the consensus on the surgical approach is undecided. A retrospective study at the Rare Cancer Center and other studies established breast-conserving surgery (BCS) as the preferred method [31,41]. Given the absence of universal treatment guidelines, Ro et al. proposed a grading system based on histological architectural patterns to select the surgical procedure. Tumors with 0 % solid component (grade 1) were treated with simple lumpectomy, tumors with 1–29 % solid component (grade 2) with mastectomy, and tumors with ≥ 30 % solid component with mastectomy and lymph node dissection [42]. This grading system is significant for treatment and prognosis, as solid or basaloid components are associated with a more aggressive clinical course [37]. In our study, 74.8 % of the patients underwent partial mastectomy, and the remainder underwent various forms of total mastectomy.

The use of non-breast-conserving surgery may be due to the

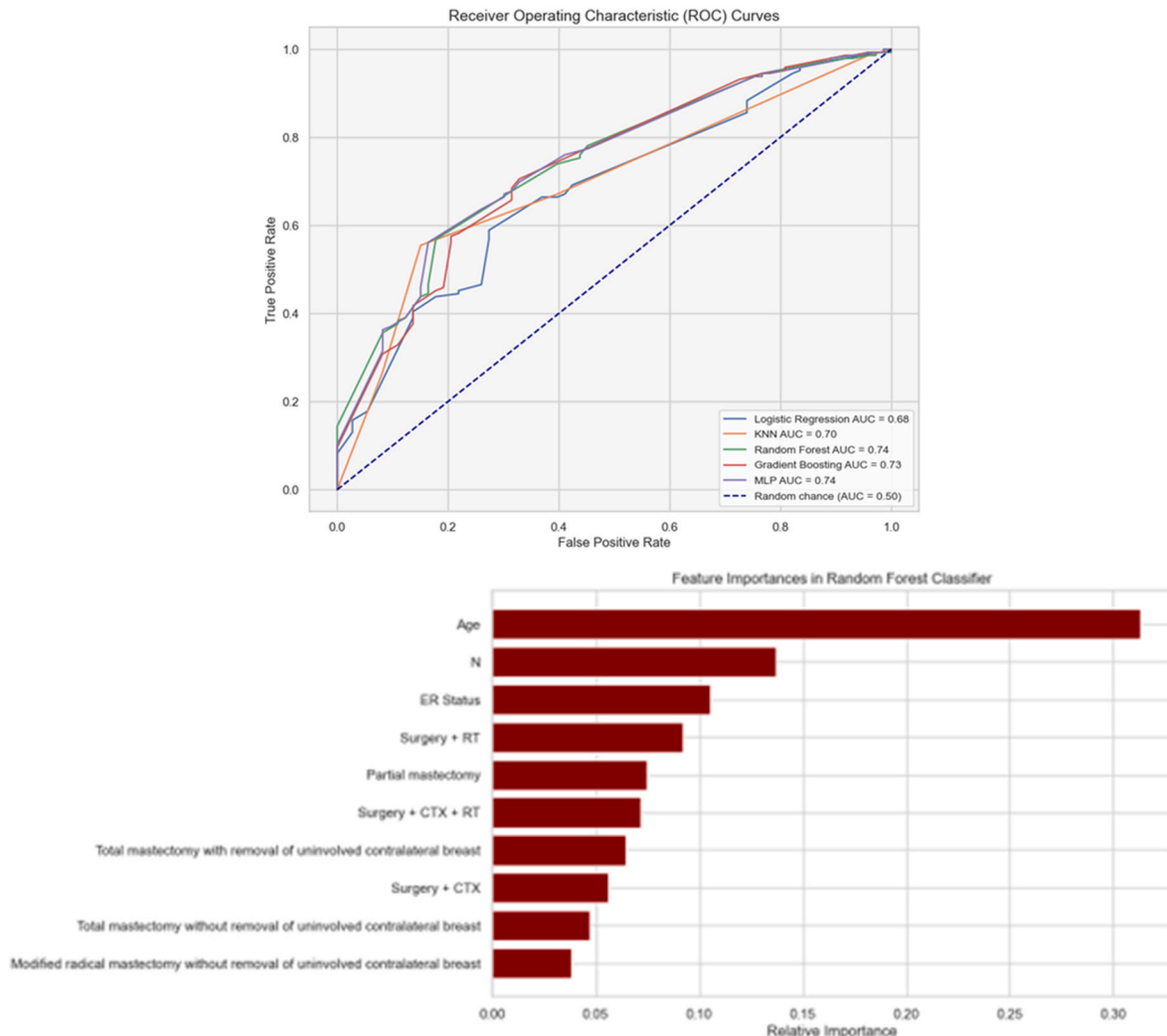


Fig. 4. A) Receiver Operating Characteristic (ROC) curves of all Machine Learning Models (MLMs); B) Permutation Features Importance (Random Forest Classifier).

perceived aggressiveness of the tumor, akin to other triple-negative breast cancers, and the increased risk of recurrence with BCS in patients not receiving postoperative radiation. Additionally, 5.4 % of patients underwent bilateral total mastectomies, likely because invasive carcinomas have been documented in the contralateral breast in BACC patients [26].

The role of RT in treating BACC remains unclear. It is typically indicated in the presence of nodal metastasis or following breast-conserving surgeries owing to increased rates of positive surgical margins, aiming to reduce recurrence risks [43]. Two studies using the SEER database reported increased OS rates among patients who received postoperative RT [44,45]. Other studies have provided supporting evidence for using postoperative RT [32,46]. A study by Khanfir et al. at the Rare Cancer Network showed that RT increased locoregional control rates by 12 % in patients who underwent lumpectomies. However, it did not increase the OS [41]. Our study found that surgery followed by RT resulted in the highest survival rates for OS and CSS. Conversely, a retrospective study over 20 years indicated that five patients treated with surgery alone remained disease-free with a median follow-up of 85 months (32). This suggests that surgery alone may suffice in selecting patients. Nonetheless, postoperative RT should be considered, especially after BCS, as it may benefit patients regardless of the margin status [41].

Adjuvant chemotherapy (CTX) is used to treat BACC, although infrequently. One study reported that only 11.3 % of patients received adjuvant CTX [39], and our findings showed that only 10.3 % of patients received CTX after surgery. This may be due to several factors, including the tumor's low grade, the indolent nature of the disease, the rarity of distant metastases/relapses, and physicians' reluctance to administer CTX due to a lack of prospective trials. Arpino et al. suggested no difference in OS rates between women who received systemic therapy and those who did not, deeming CTX unnecessary given its side effects and costs. Similar findings are reported in the literature [36,47,48]. In our study, adjuvant CTX was associated with lower OS rates than surgery alone or surgery + RT. This may be unrelated to CTX itself, as patients with BACC receiving CTX generally have more advanced disease stages, more aggressive forms of BACC, and local/distant metastases. Current NCCN guidelines recommend adjuvant CTX for node-positive patients and consider it for those with high-grade lesions, tumors >3 cm, or metastatic disease [49,50]. One study indicated that distant recurrence rates were not negligible, suggesting the use of adjuvant CTX in some cases [51]. Different reports on the efficacy of CTX in solid-basaloid subtype cases have also been reported [52,53]. However, the use of adjuvant CTX therapy remains controversial.

The strengths of this study include its comprehensive data collection, ensuring a large and diverse sample size, and robust methodology using various statistical tools and ML algorithms to analyze prognostic factors and treatment outcomes. An innovative approach to developing a practical web-based tool for clinicians is notable. However, the retrospective design of this study may have introduced potential bias, and the absence of tumor grade and detailed data on radiotherapy and chemotherapy dosages are significant limitations.

5. Conclusion

Age, metastatic involvement of the lymph nodes, and ER status are the most important prognostic factors for survival in patients with BACC. Postoperative RT improves the overall and cancer-specific survival rates, particularly after breast-conserving surgery. There is no need to use adjuvant CTX as its efficacy remains controversial and may not significantly improve survival outcomes, allowing us to avoid associated side effects. In addition, our innovative web-based tool offers clinicians a practical resource for predicting patient survival and effectively tailoring treatment plans.

CRedit authorship contribution statement

Sakhr Alshwayyat: Software, Formal analysis, Conceptualization. **Mahmoud Bashar Abu Al Hawa:** Writing – original draft, Resources. **Mustafa Alshwayyat:** Software, Formal analysis. **Tala Abdulsalam Alshwayyat:** Writing – original draft, Methodology. **Siya sawan:** Writing – original draft, Resources. **Ghaith Heilat:** Writing – review & editing, Supervision. **Hanan M. Hammouri:** Methodology, Formal analysis. **Sara Mheid:** Writing – review & editing, Supervision. **Batool Al Shweiat:** Writing – review & editing, Resources. **Hamdah Hanifa:** Writing – review & editing, Resources.

Ethics approval and consent to participate

The ethical approval and informed consent statements are not applicable for this type of research. Authorization and data were obtained through the SEER website and database, respectively.

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Conflicts of interest

None declared.

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